Untangling the Connection between Caloric Restriction and Prolonged Survival in Non-Human Primates: A Multi-omics Approach

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Background

Age is a major risk factor for a variety of chronic diseases. Caloric restriction (CR) has been proven to alleviate risk factors accompanying aging and improve longevity in several model organisms.

A 30-year longitudinal study addressing CR in rhesus macaques, a close living relative to *Homo Sapiens*, revealed that animals with 30% CR had 2.9 fewer morbidity incidences of various age-related pathologies (Mattison 2017). However, little is known about the changes in circulating factors impacting survival in non-human primates (NHP).

By integrating metabolomics, lipidomics and proteomics, we can provide a network of biomarkers linked to the benefits of CR, including longevity and delayed disease onset.

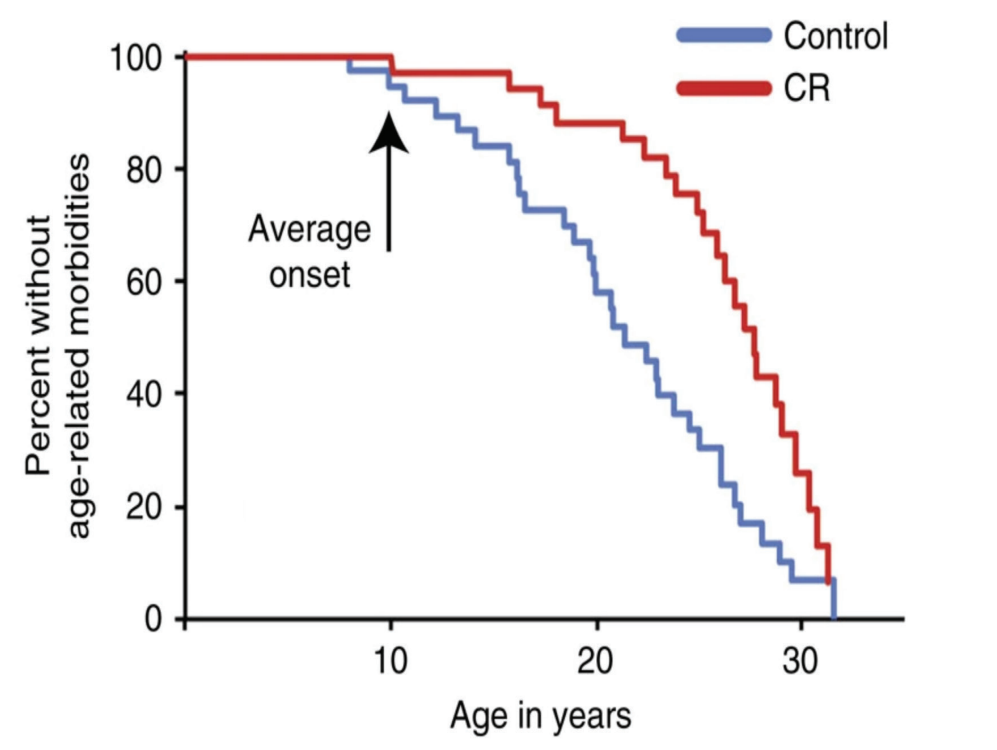


Fig.1 Age-related mortality in non-human primates.

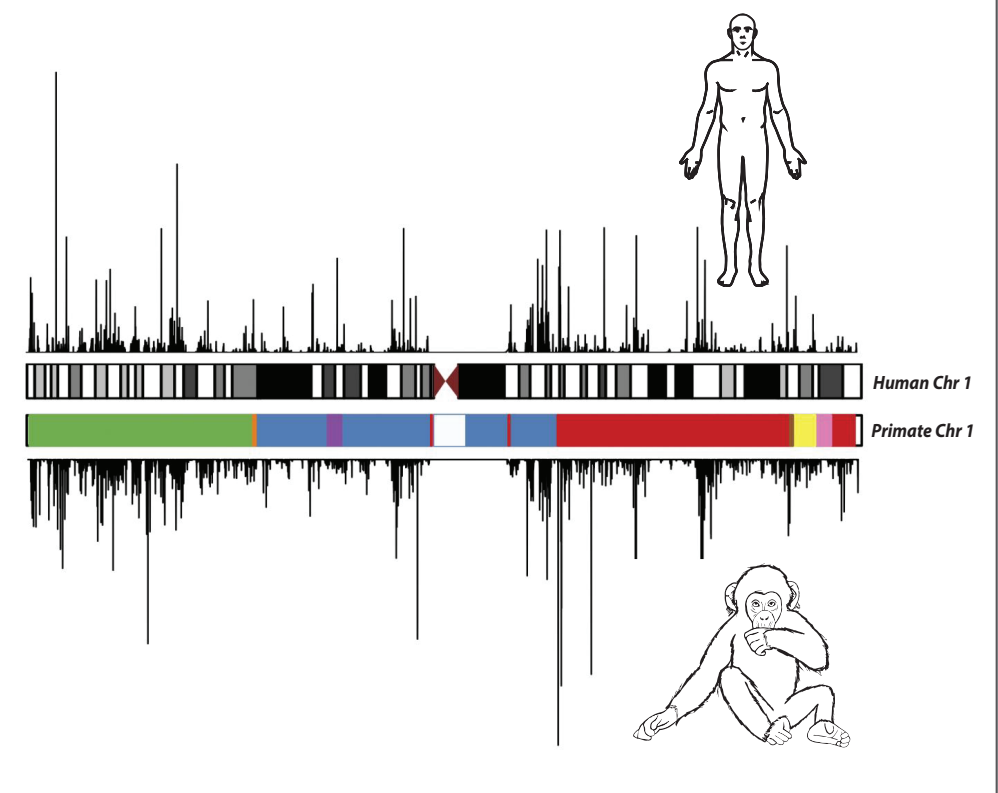
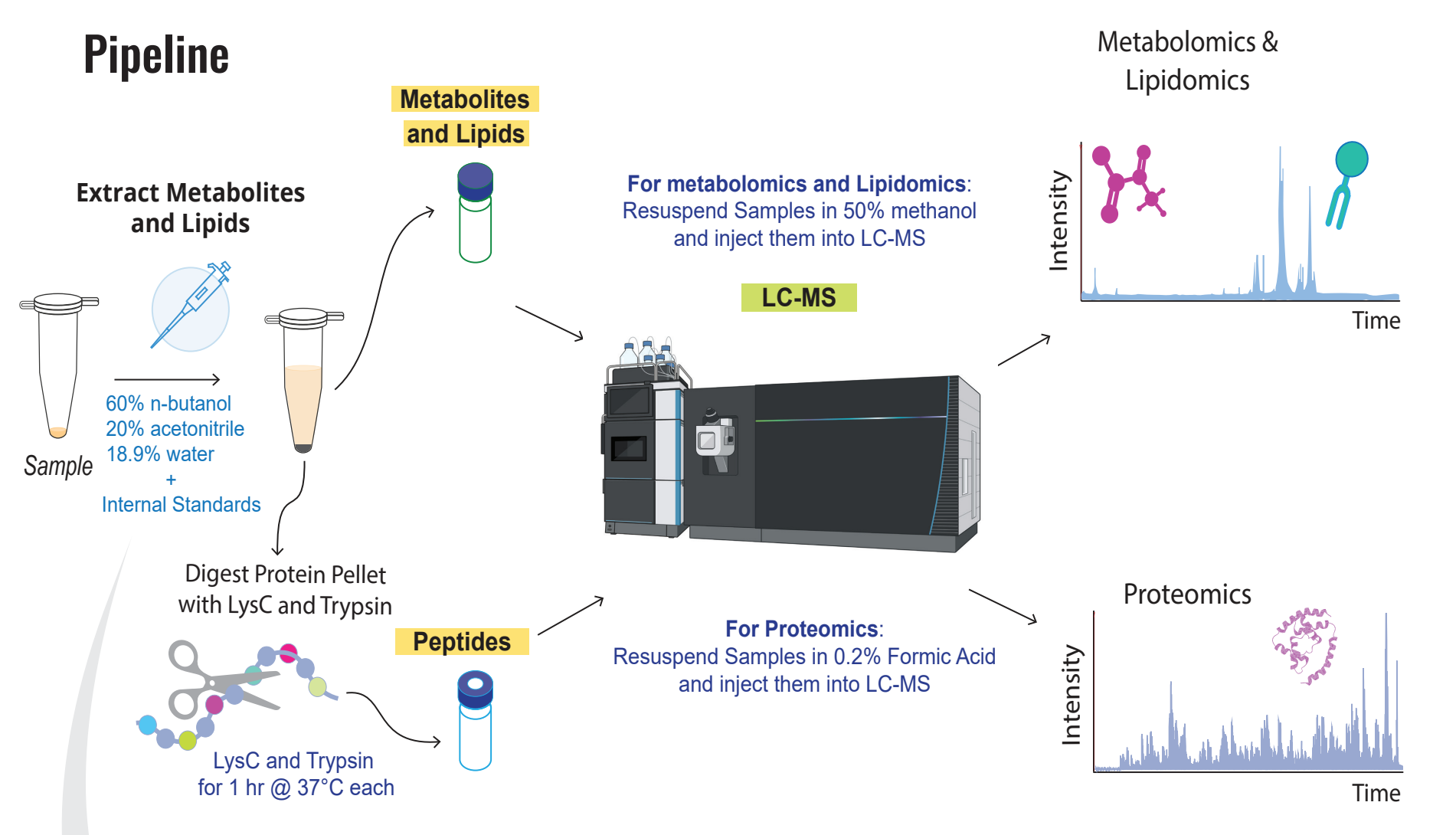


Fig.2 High genetic similarity between Primates and *Homo Sapiens*.

Workflow


Pipeline
Extract Metabolites and Lipids
Sample
60% n-butanol
20% acetonitrile
18.9% water
+
Internal Standards
Metabolites and Lipids
Digest Protein Pellet with LysC and Trypsin
LysC and Trypsin for 1 hr @ 37°C each
Peptides
For metabolomics and Lipidomics: Resuspend Samples in 50% methanol and inject them into LC-MS
LC-MS
For Proteomics: Resuspend Samples in 0.2% Formic Acid and inject them into LC-MS
Metabolomics & Lipidomics
Proteomics
Intensity
Time

Muehlbauer, L. et. al., Analytical Chemistry (2023)

Internal Standards We incorporated SPLASH LIPIDOMIX, a mixture of stable isotope-labeled lipid standards, and C^{13} N^{15} amino acids, to monitor and correct for variations due to sample preparation.

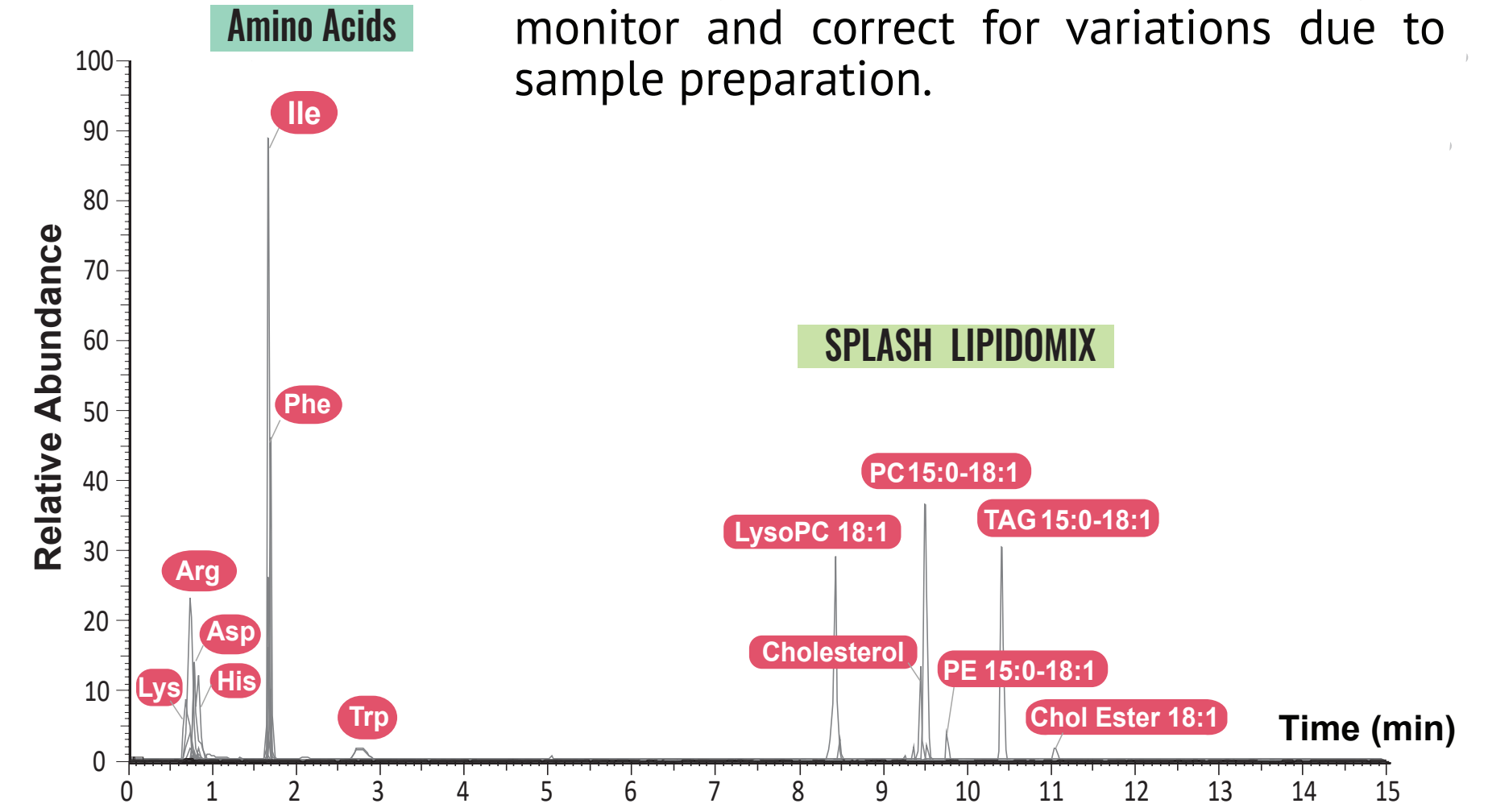
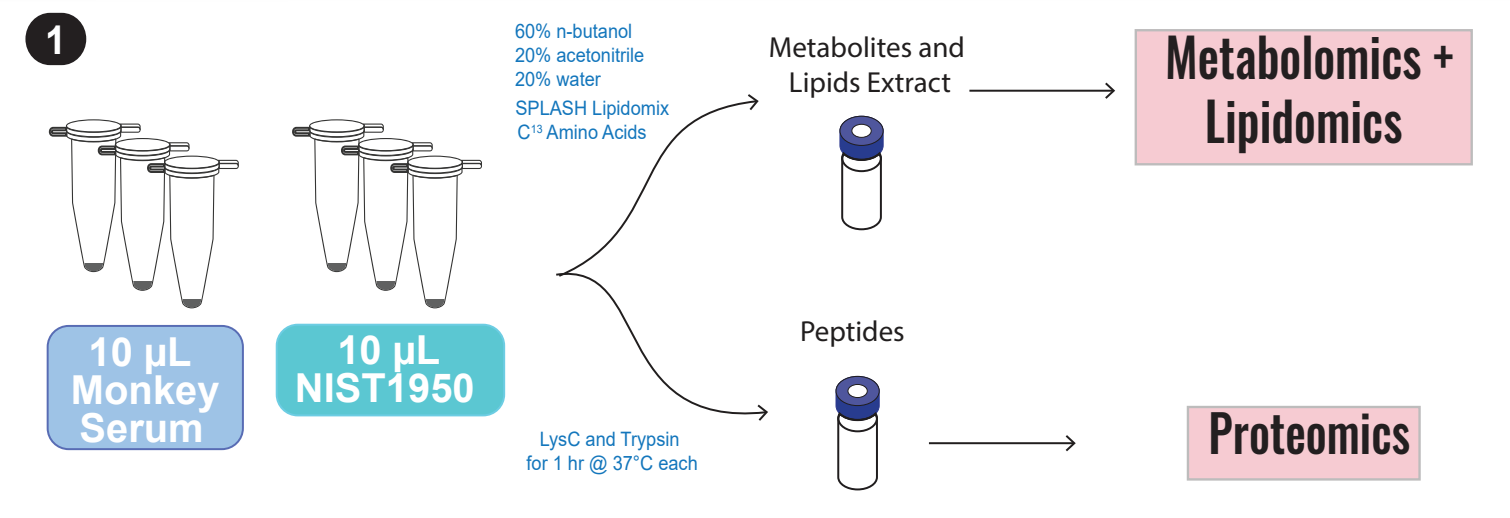


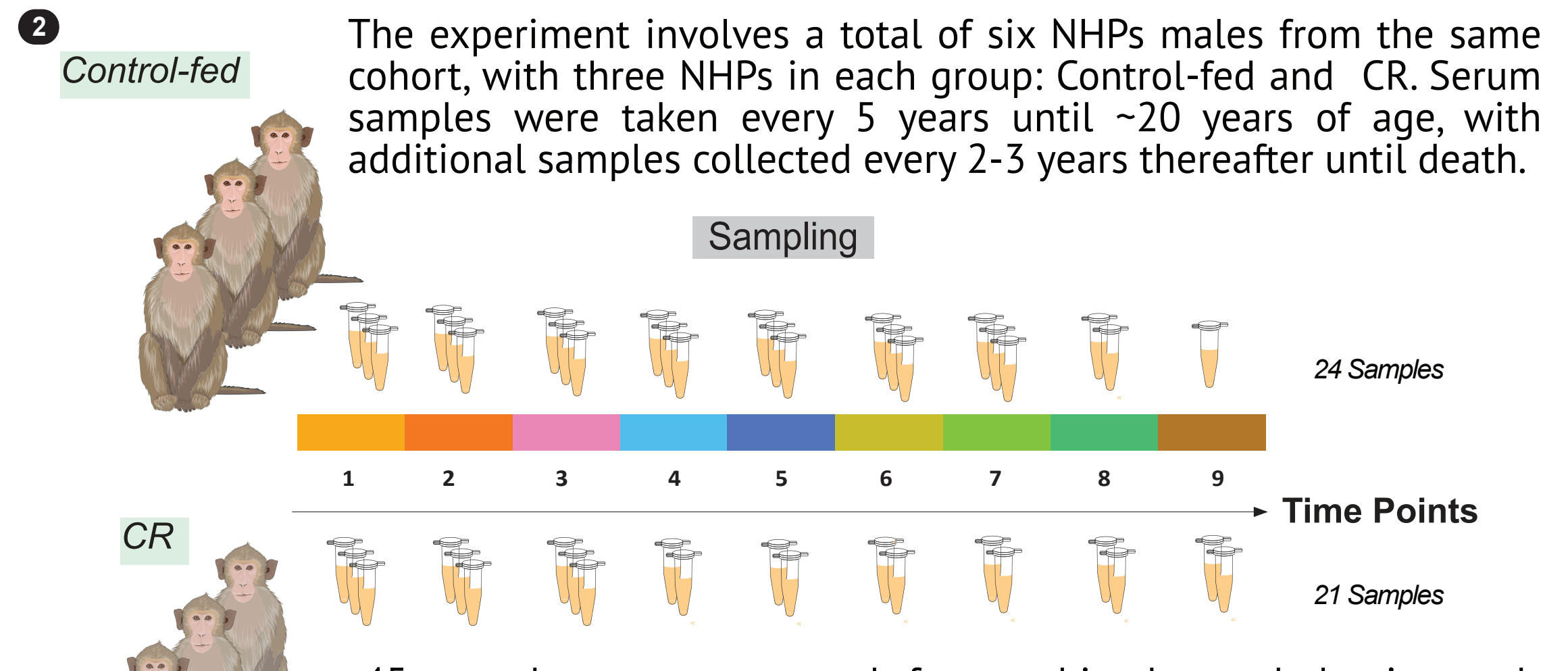
Fig.3 Distinct peaks corresponding to SPLASH lipids and C13-labeled amino acids in our samples in the positive mode.

Methods and Experiments



To test our methods for multi-omics analyses, we did a combined metabolomics and lipidomics, and a separate proteomics runs to NIST1950 and Pooled Monkey Serum (3 samples, each).

The experiment involves a total of six NHPs males from the same cohort, with three NHPs in each group: Control-fed and CR. Serum samples were taken every 5 years until ~20 years of age, with additional samples collected every 2-3 years thereafter until death.



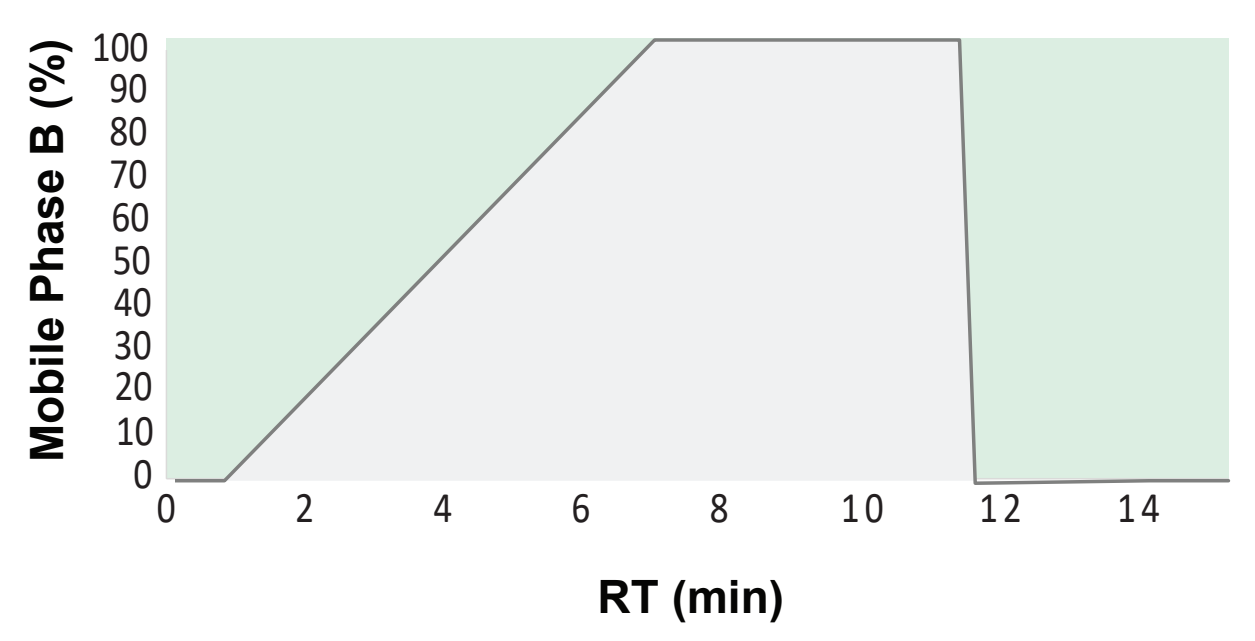
45 samples were prepared for combined metabolomics and lipidomics analysis, including 10 quality control (QC) samples from pooled monkey serum. Metabolites and lipids were extracted from 10 µL of serum using 500 µL of 3:1:1 n-butanol:acetonitrile:water, with addition of internal standards SPLASH (Avanti, 10 µL/mL final) and C^{13} N^{15} amino acid (Sigma, 1 µL/mL final).

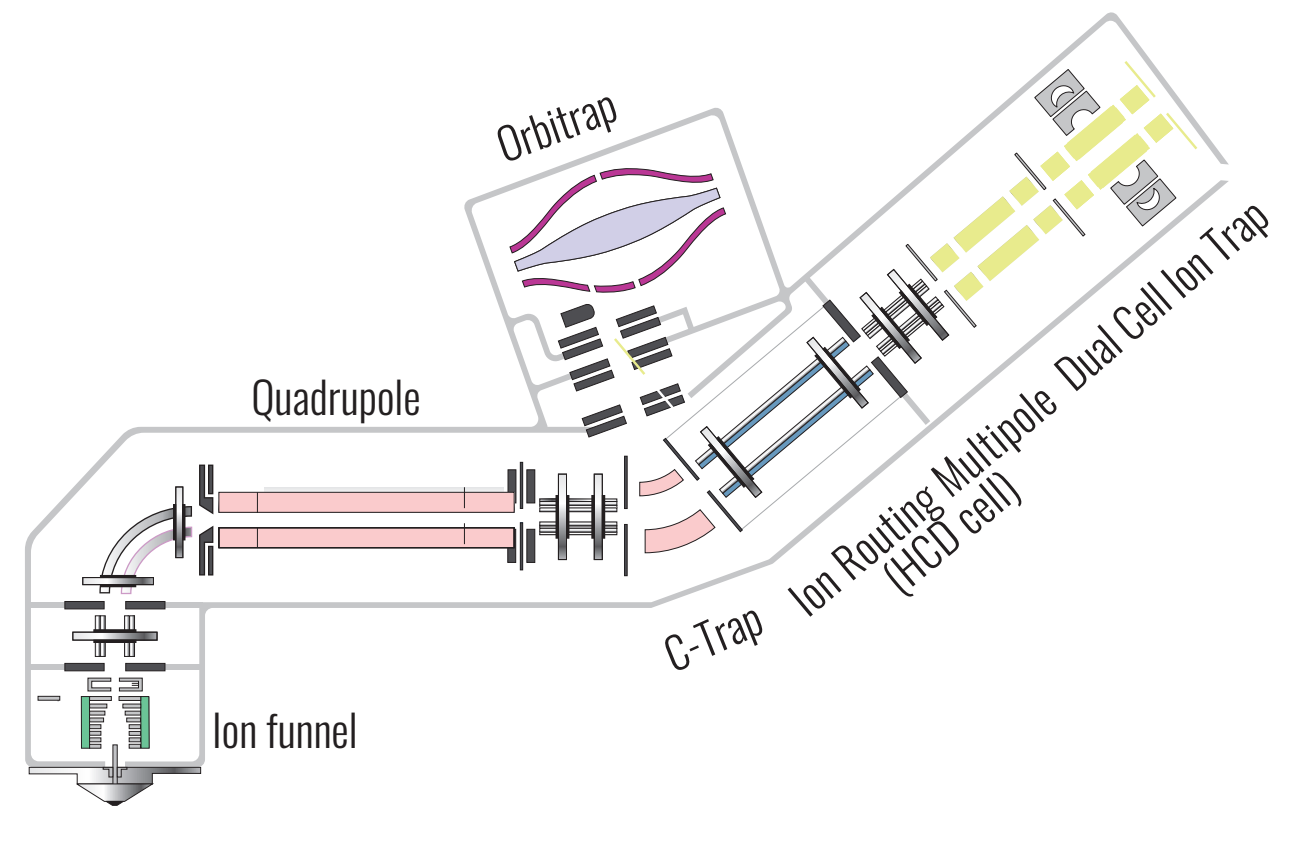
For metabolomics and lipidomics, experiments were performed on a Vanquish HPLC coupled to a Orbitrap Eclipse Tribrid MS system (Thermo Fisher). Separation was accomplished using a gradient of 0-100% mobile phase B across 6 min. Analytes were separated using an ACQUITY HSS T3 C18, 1 mm × 100 mm, 1.8 µm column (Waters). The Mobile phases consisted of 0.2% formic acid (FA) in water (A), and 0.2% formic acid, 5mM Ammonium formate (AF) in (9:1) Isoprpropanol (IPA) /Acetonitrile (ACN) (B). Data were acquired with polarity switching with data-dependent Orbitrap MS2.

For Proteomics, experiments were performed on a Neo Vanquish HPLC coupled to a Orbitrap Eclipse Tribrid MS system (Thermo Fisher). Separation was accomplished using a gradient of 0-50% mobile phase B across 71 min. Analytes were separated using 75 µm inner diameter column, ~30 cm long, with 1.7 µm, 130 Å pore size BEH-C18 particles packed in-house (Waters). The mobile phases consisted of 0.2% FA in water (A), and 80% ACN and 0.2% FA in water (B). Data were acquired in positive mode with data-dependent MS2 collected in the linear iontrap with "turbo" mode.

Data were processed with Compound Discoverer 3.3 (Thermo), LipiDex (Hutchins 2018), and MaxQuant (Cox 2008).

Metabolomics and Lipidpomics Gradient

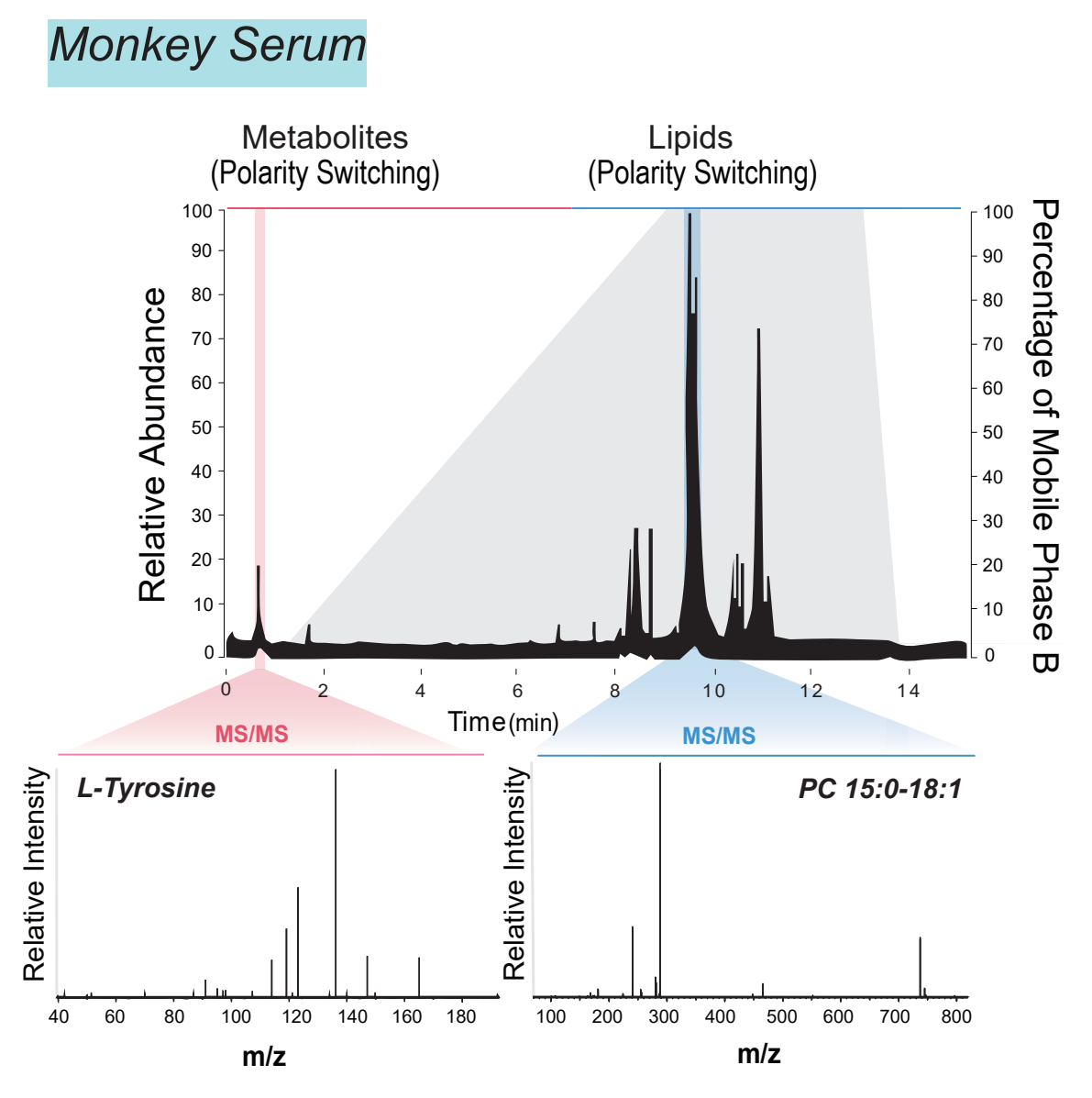


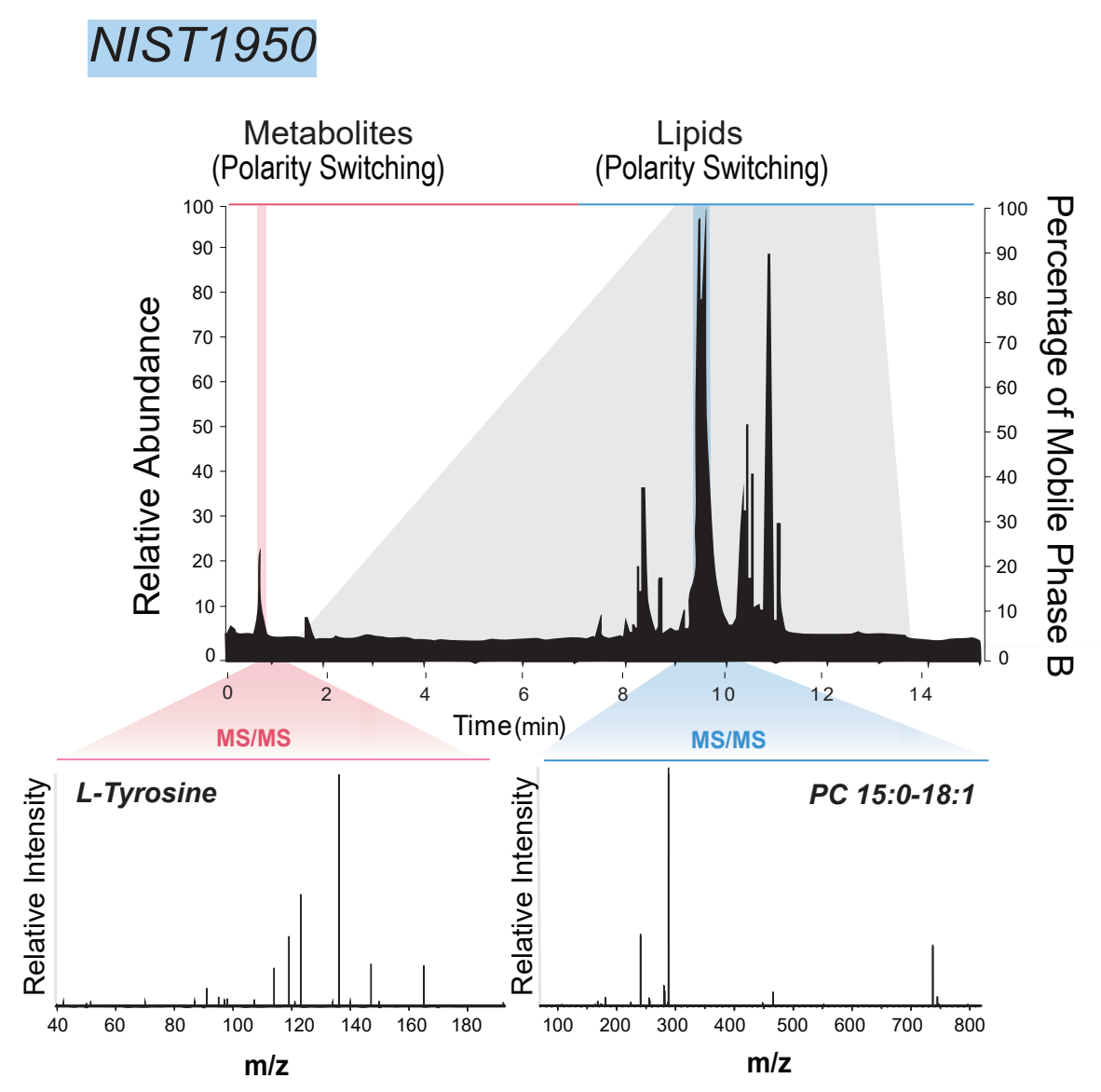


Orbitrap Eclipse Instrument

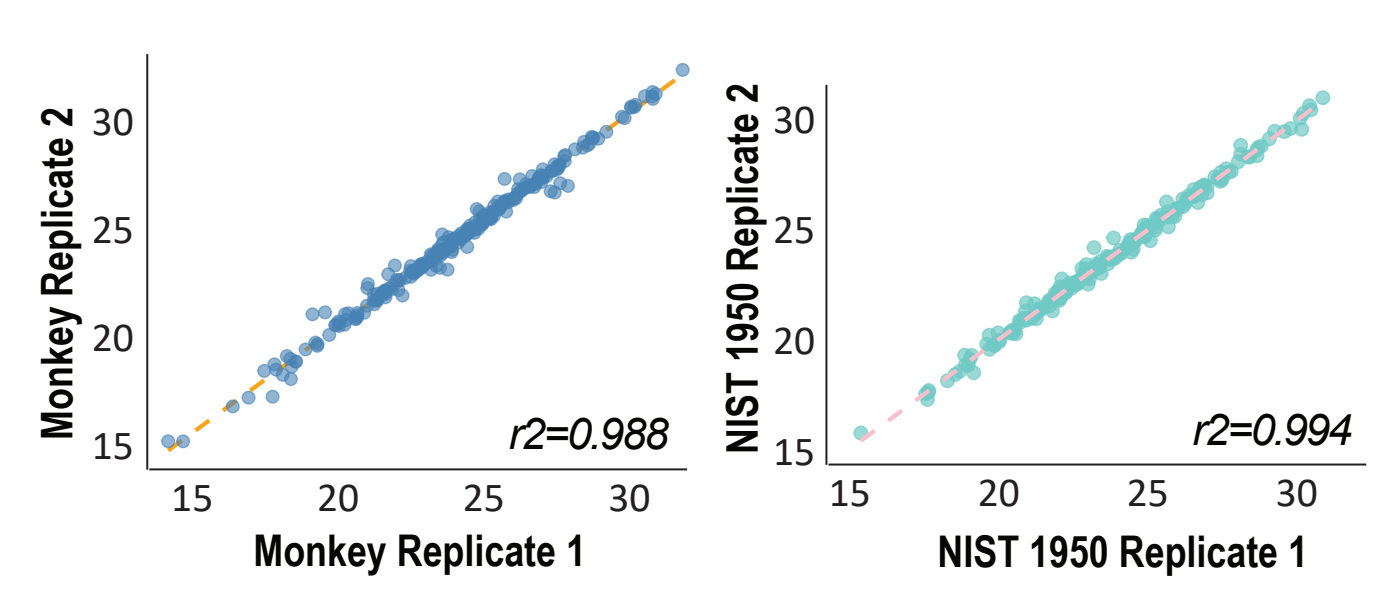
Testing Method: Monkey Serum and Human NIST 1950

Metabolomics + Lipidomics

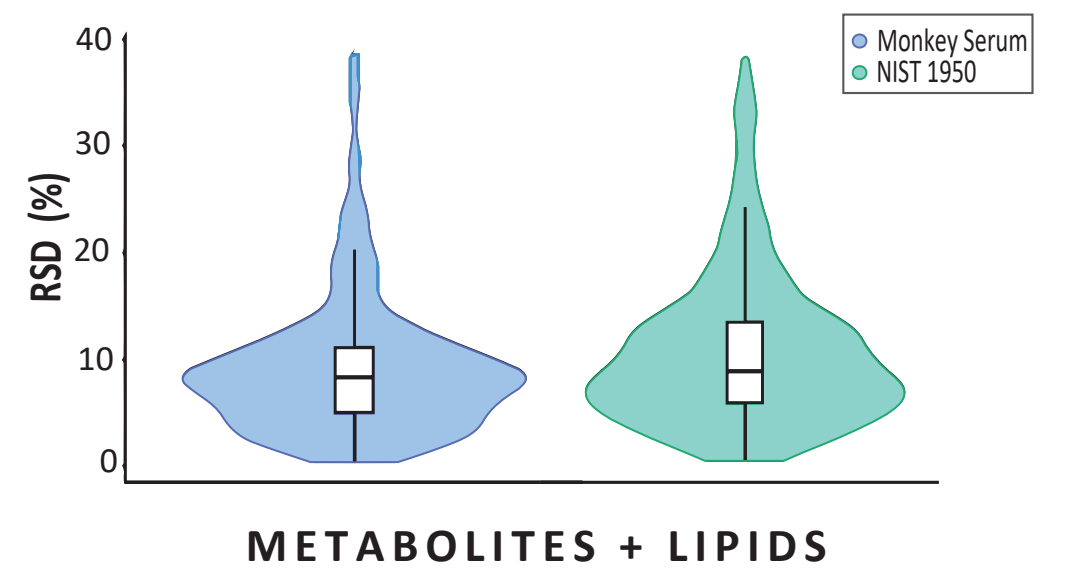




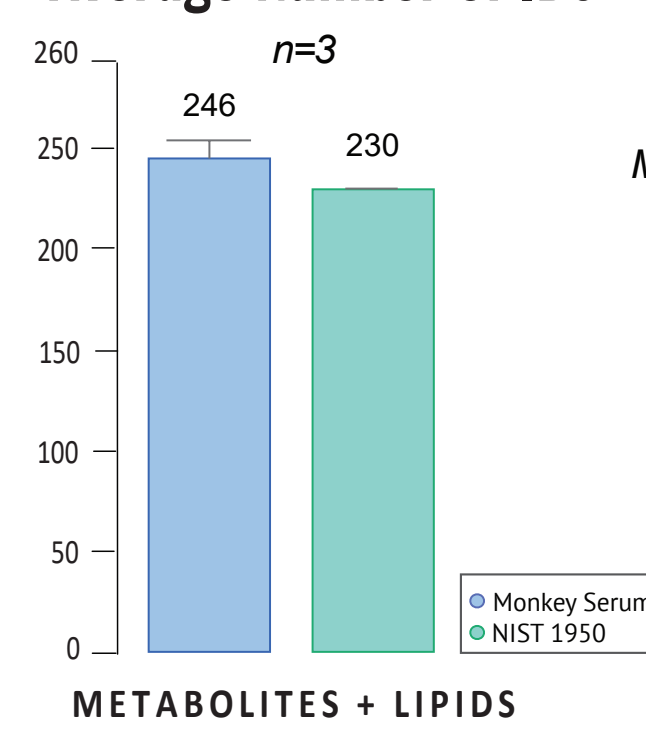
Technical Replicates



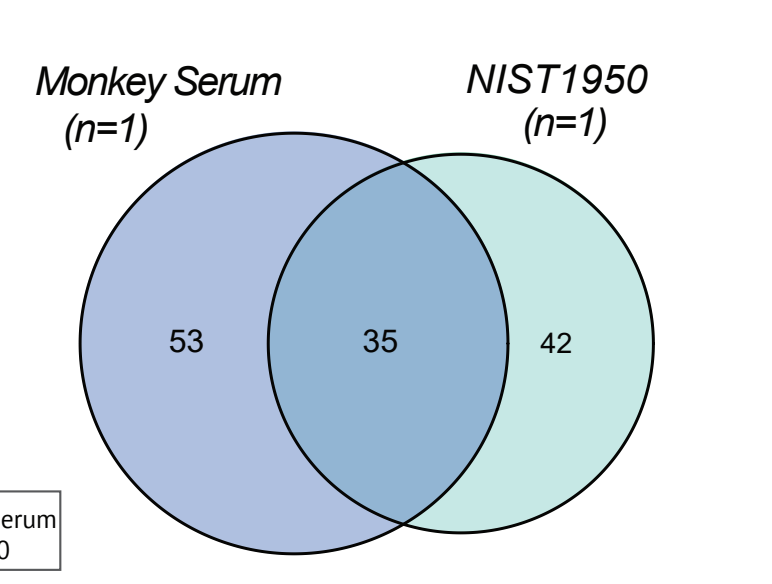
Distribution of RSDs



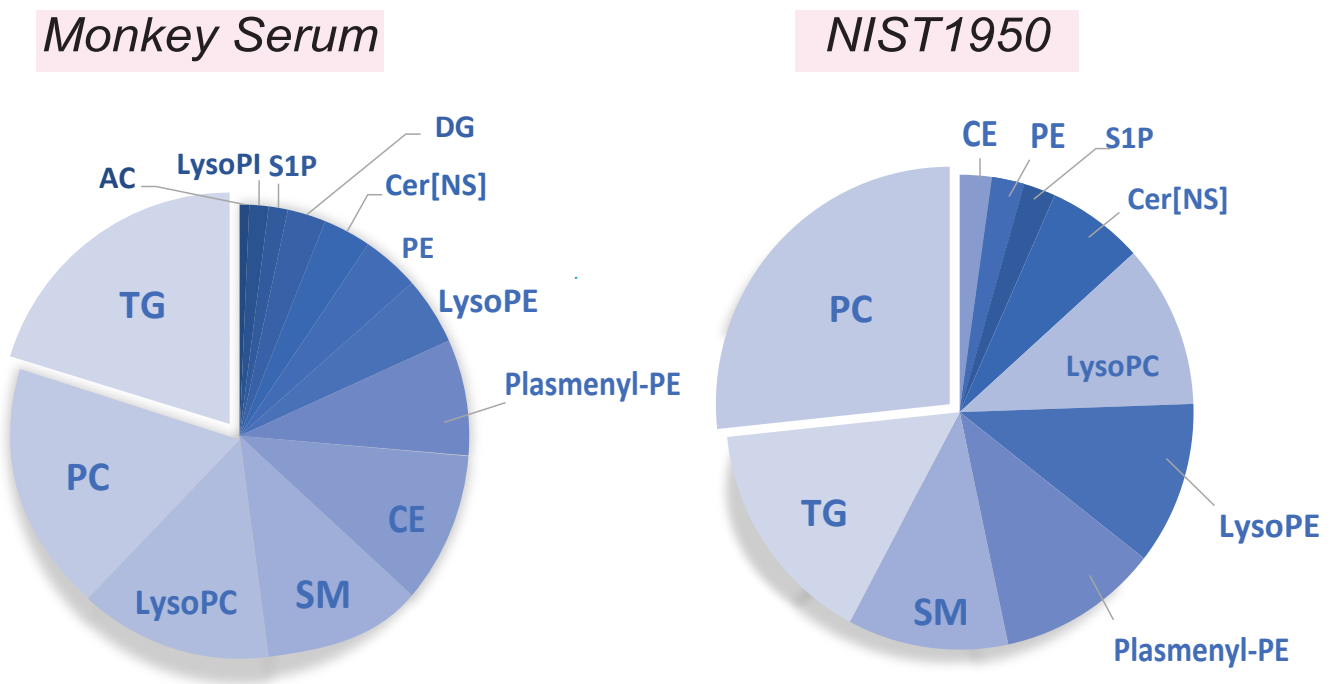
Average Number of IDs



Unique and Common IDs

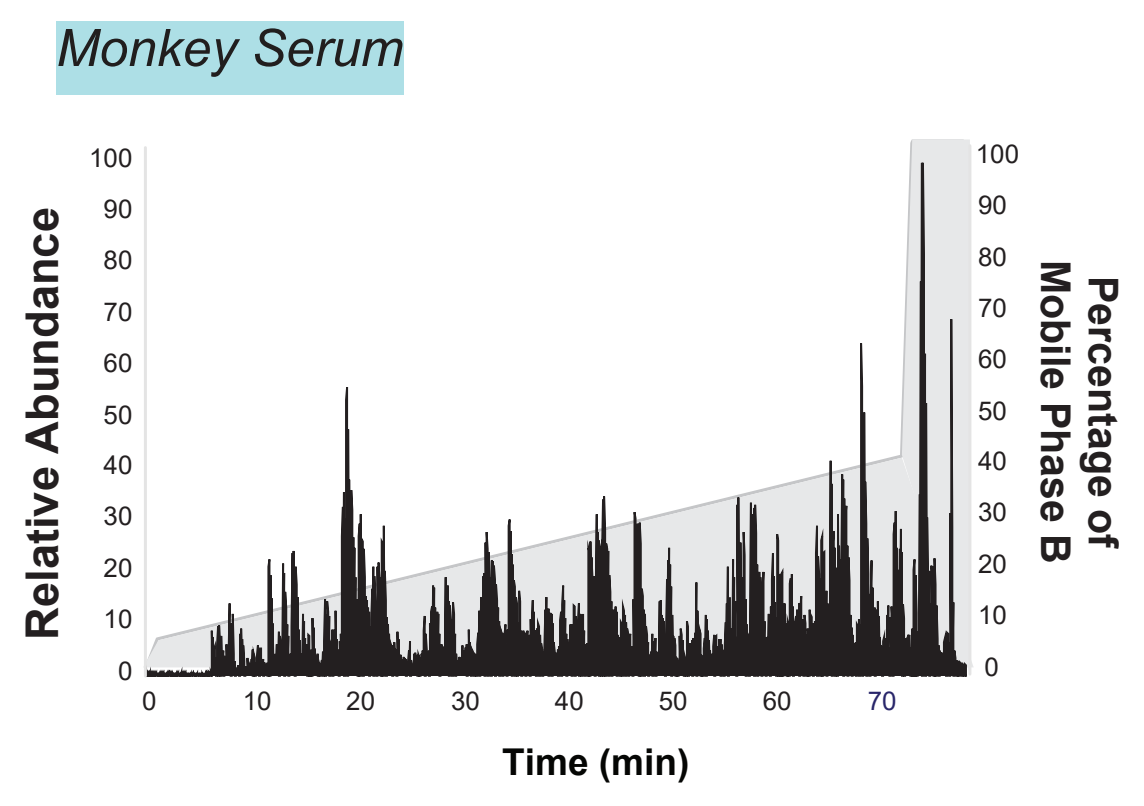


Lipid Composition

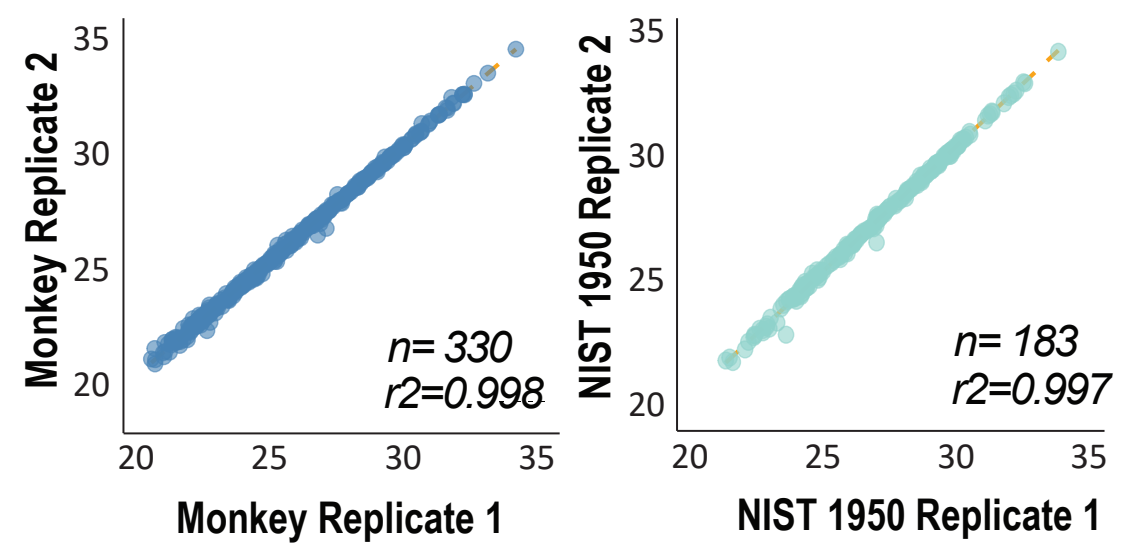


- Our method demonstrates good reproducibility in both NIST 1950 and pooled monkey serum samples, as evidenced by the highly similar results of technical replicates analysis and the low variability of distribution of relative standard deviation (RSD).
- The averaged metabolite and lipid yields were found to be higher in monkey serum compared to NIST 1950 samples. Surprisingly, we observed a lower degree of commonality in the identified metabolite and lipid between the two sample types.
- A slight difference in lipid composition was observed: While phosphatidylcholines (PC) was the most dominant lipid class in NIST 1950, triglycerides (TGs) exhibited the highest abundance in monkey serum.

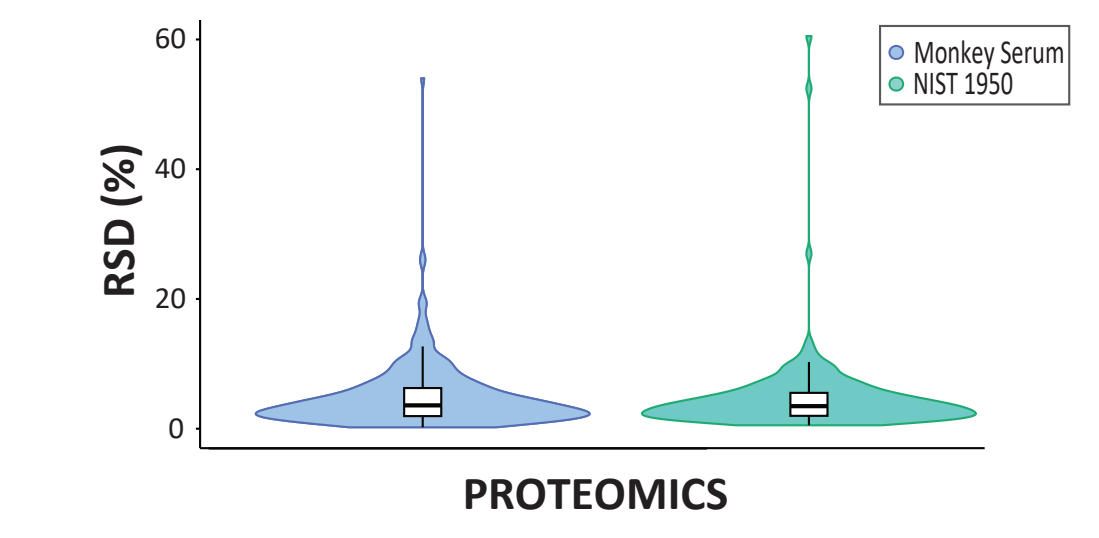
Proteomics



Technical Replicates



Distribution of RSDs



- The averaged number of identified protein groups in monkey serum and NIST 1950 are 378 and 198, respectively.

Metabolites and Lipids Linked to Lifespan

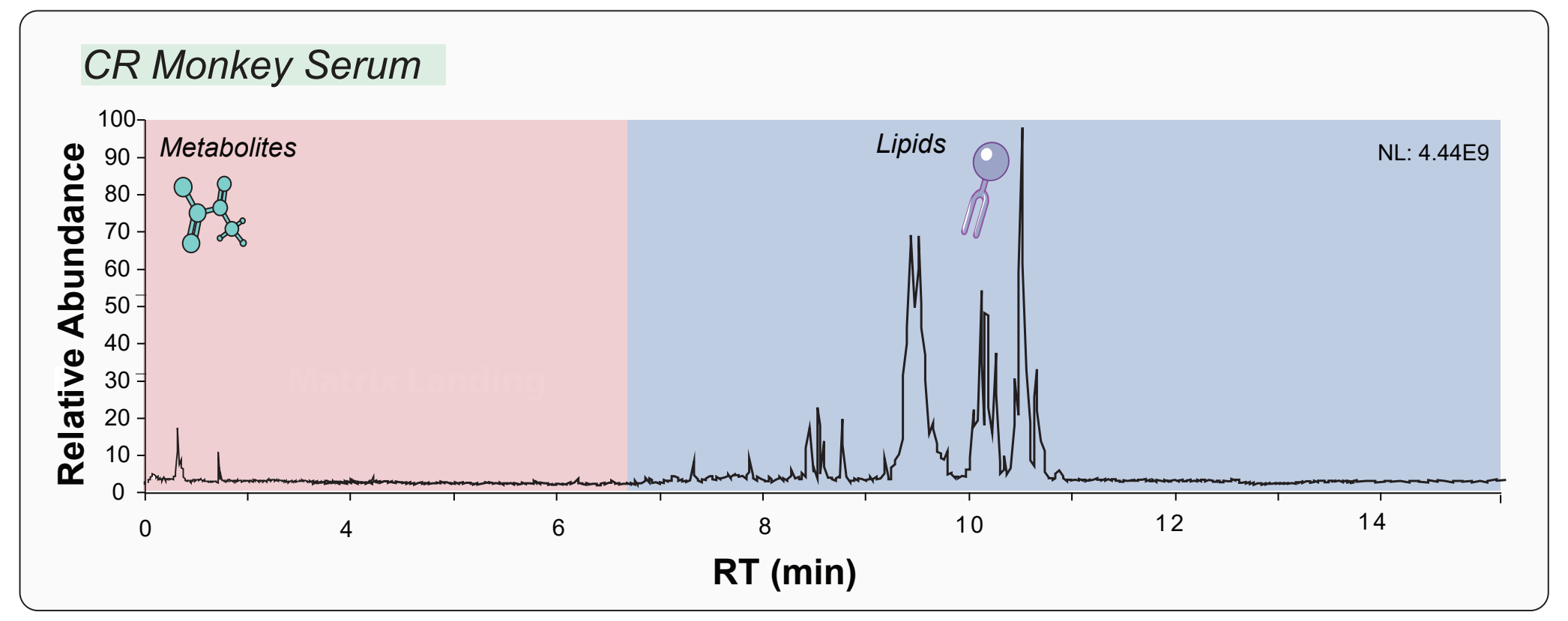
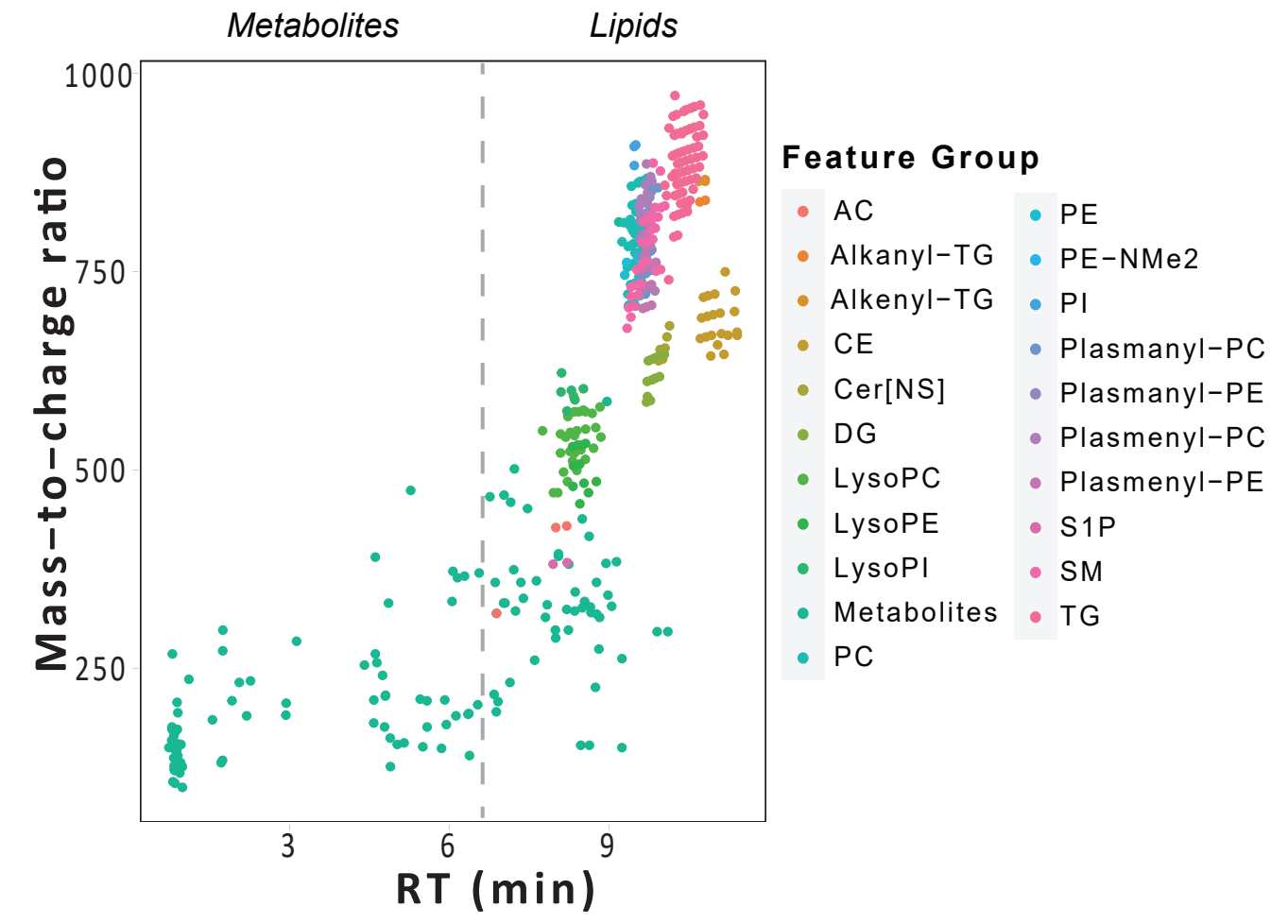
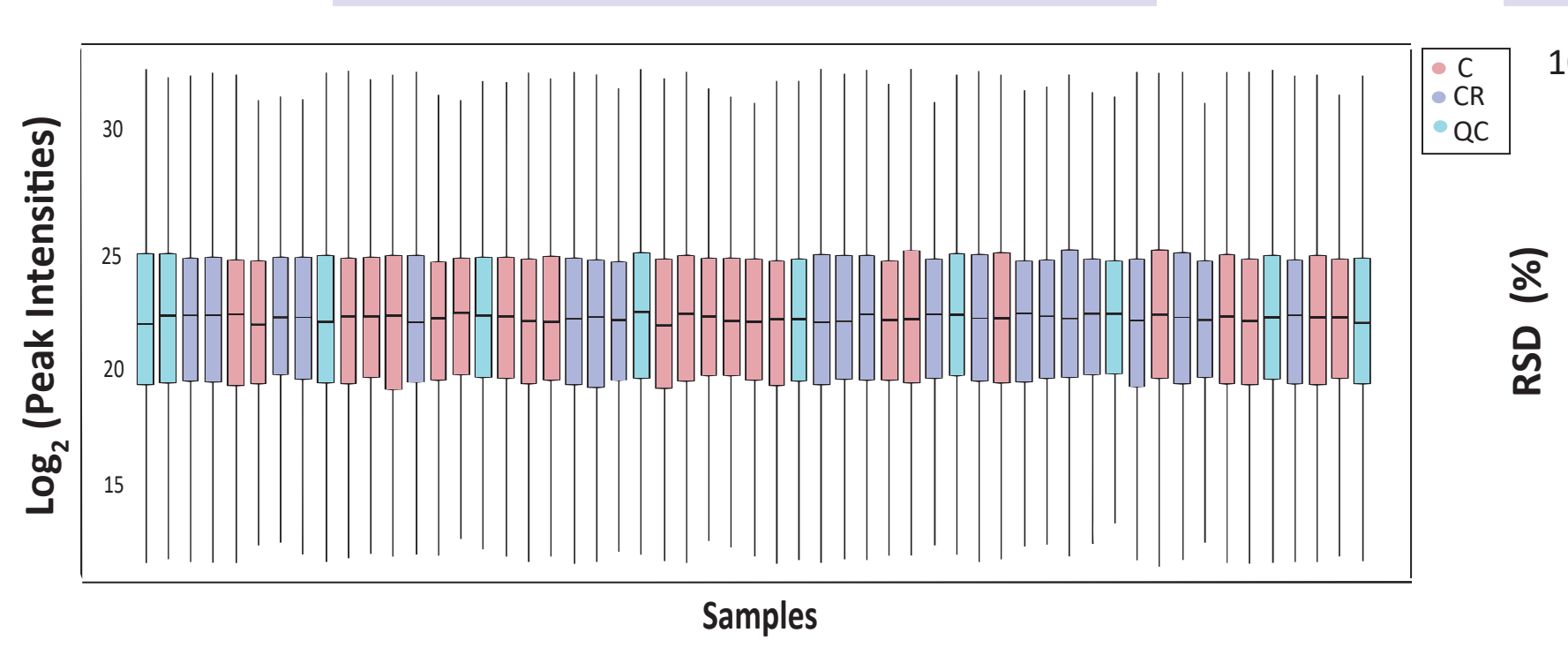


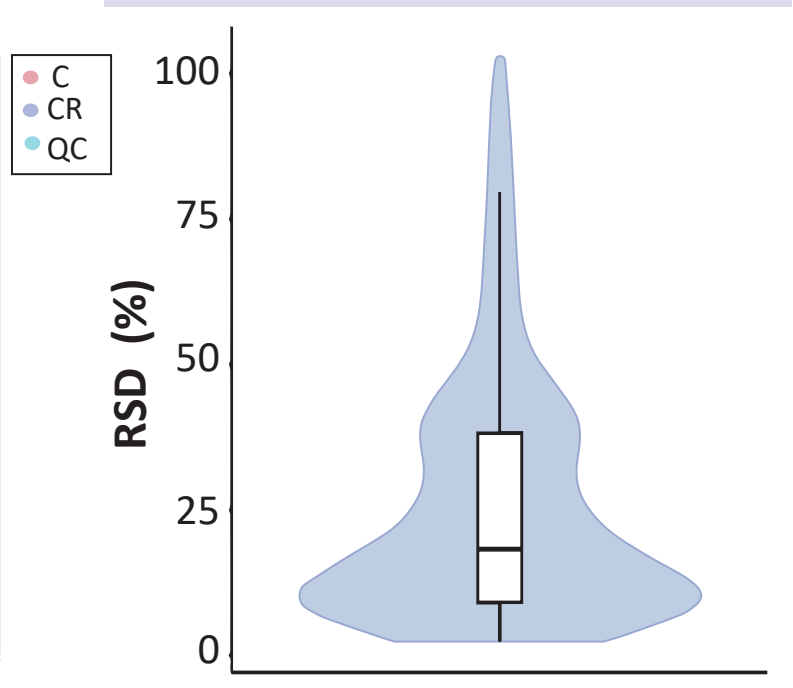
Fig.4 Distinct peaks corresponding to SPLASH lipids and C13-labeled amino acids in our samples in positive mode.



Log2 Peak Intensities of Batch 1

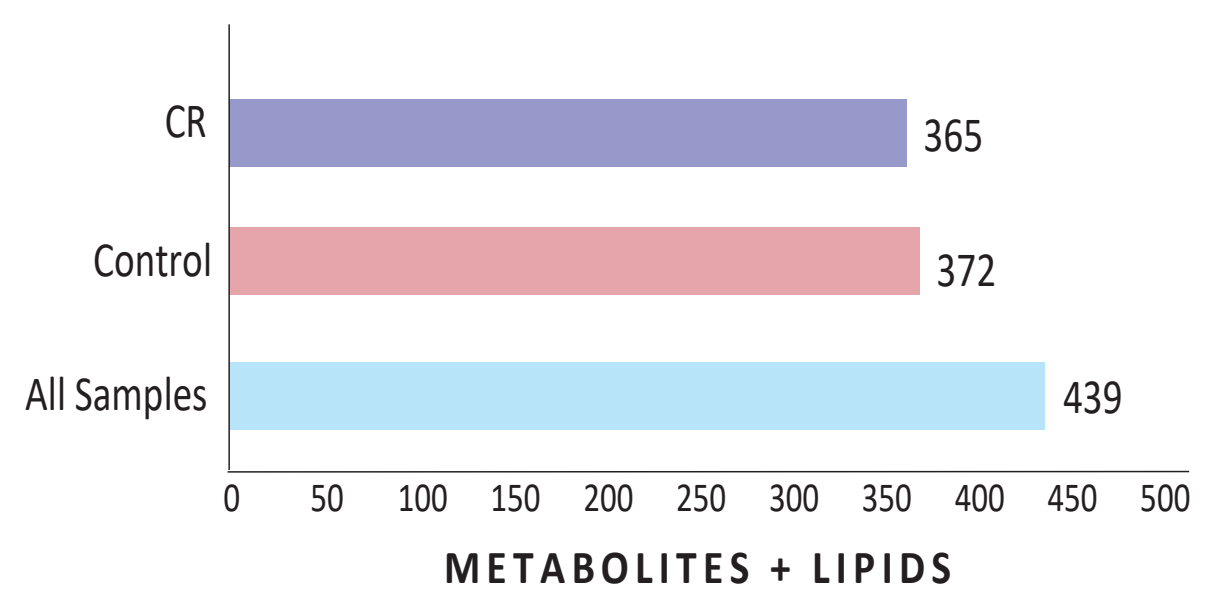


Distribution of RSD in QC Samples

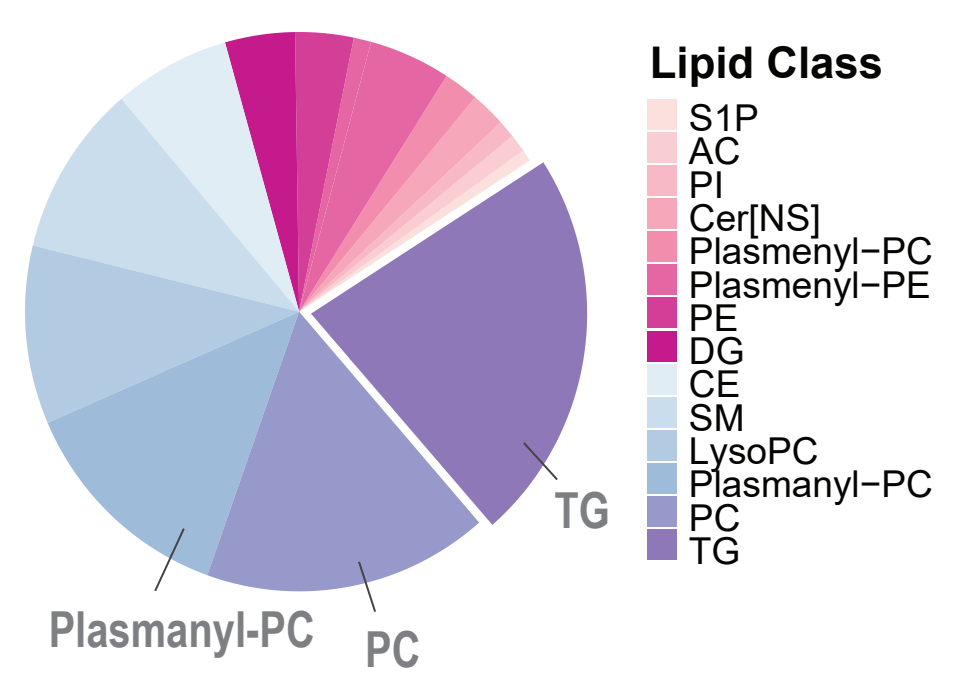


- Retention time vs m/z for all features exhibits a pattern of linearity, with features from the same group clustered together, indicating a consistent elution behavior.
- The RSD analysis of QC samples and the distribution of peak intensities among samples demonstrated good instrument and method performance.

Number of Identified Metabolites and Lipids

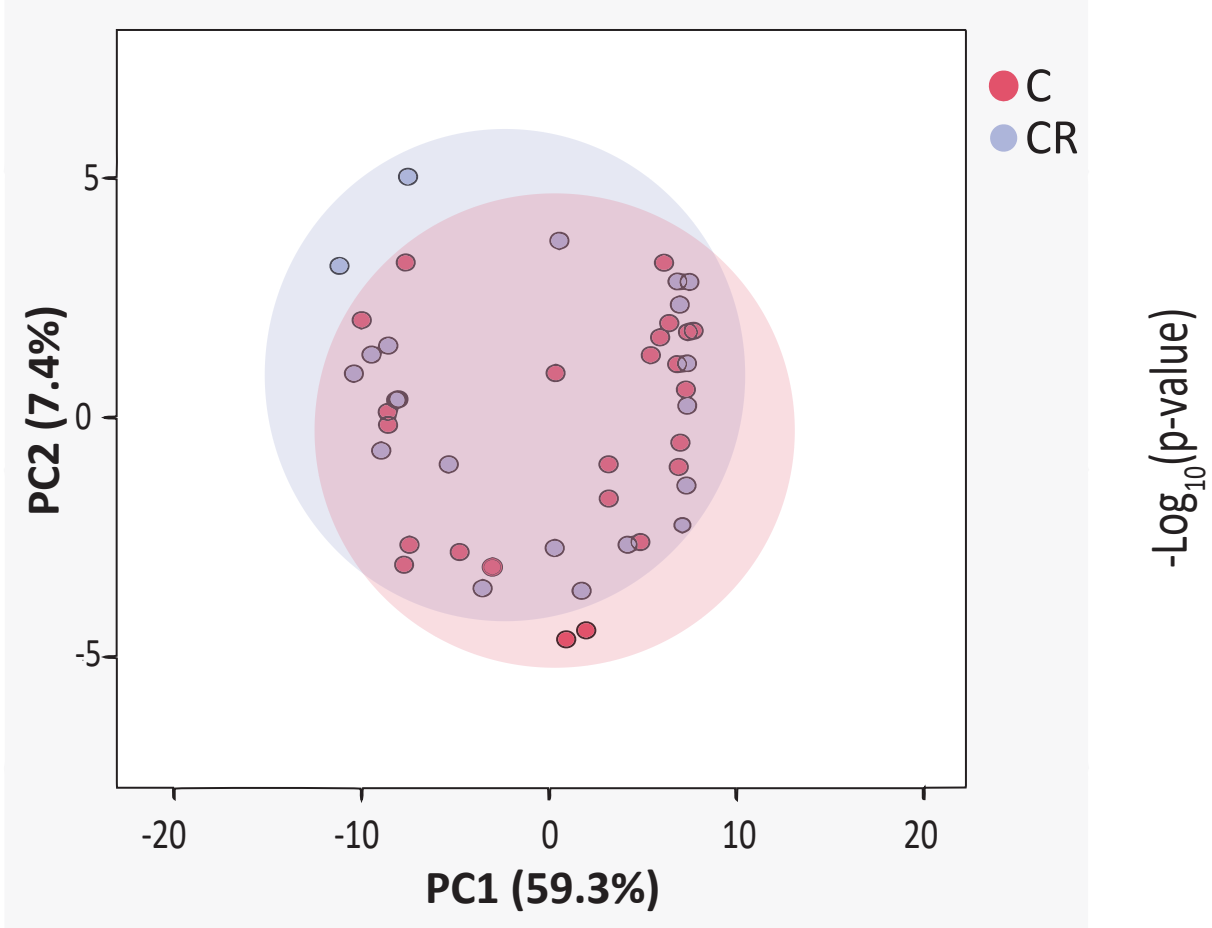


Lipid Composition

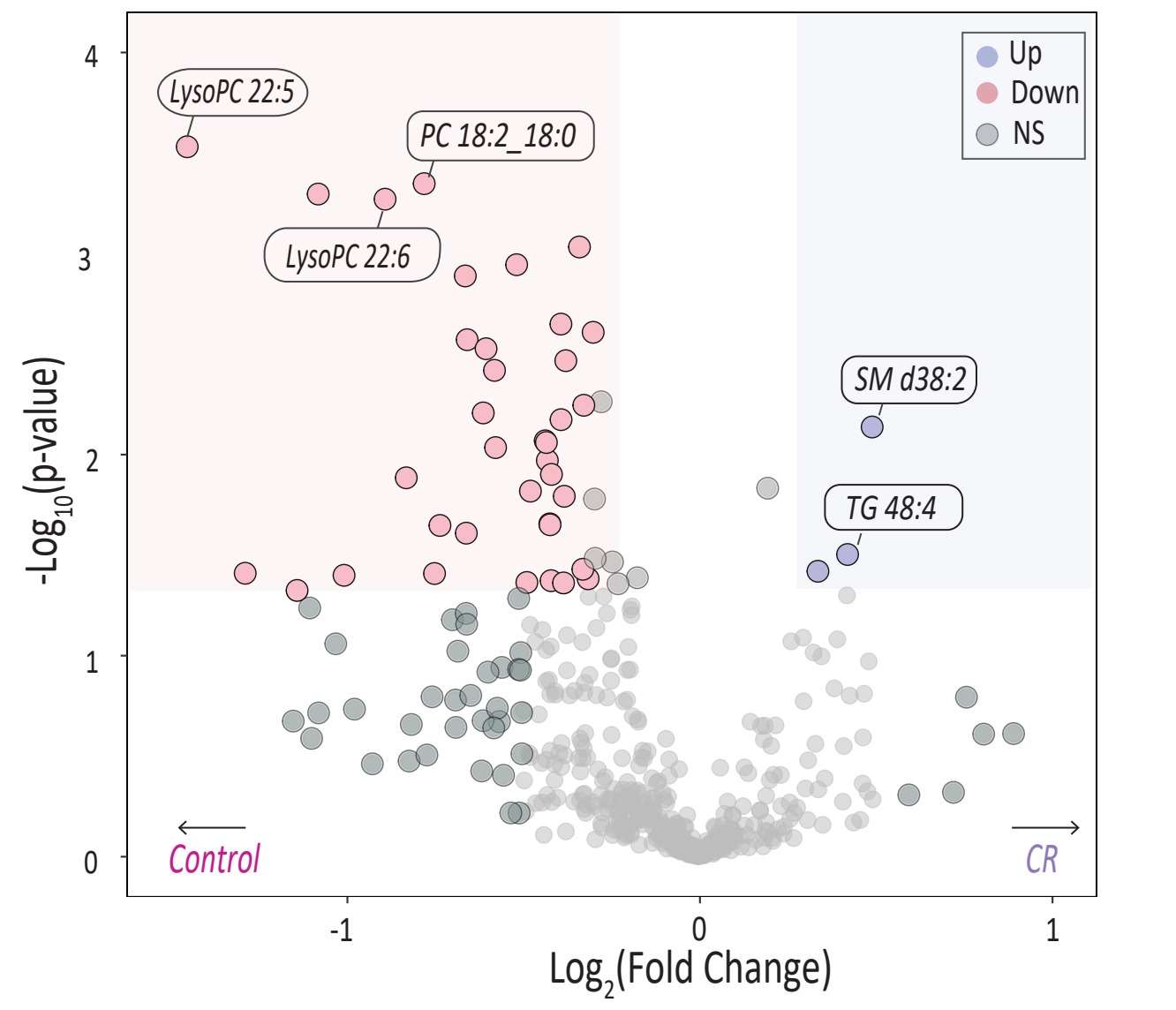


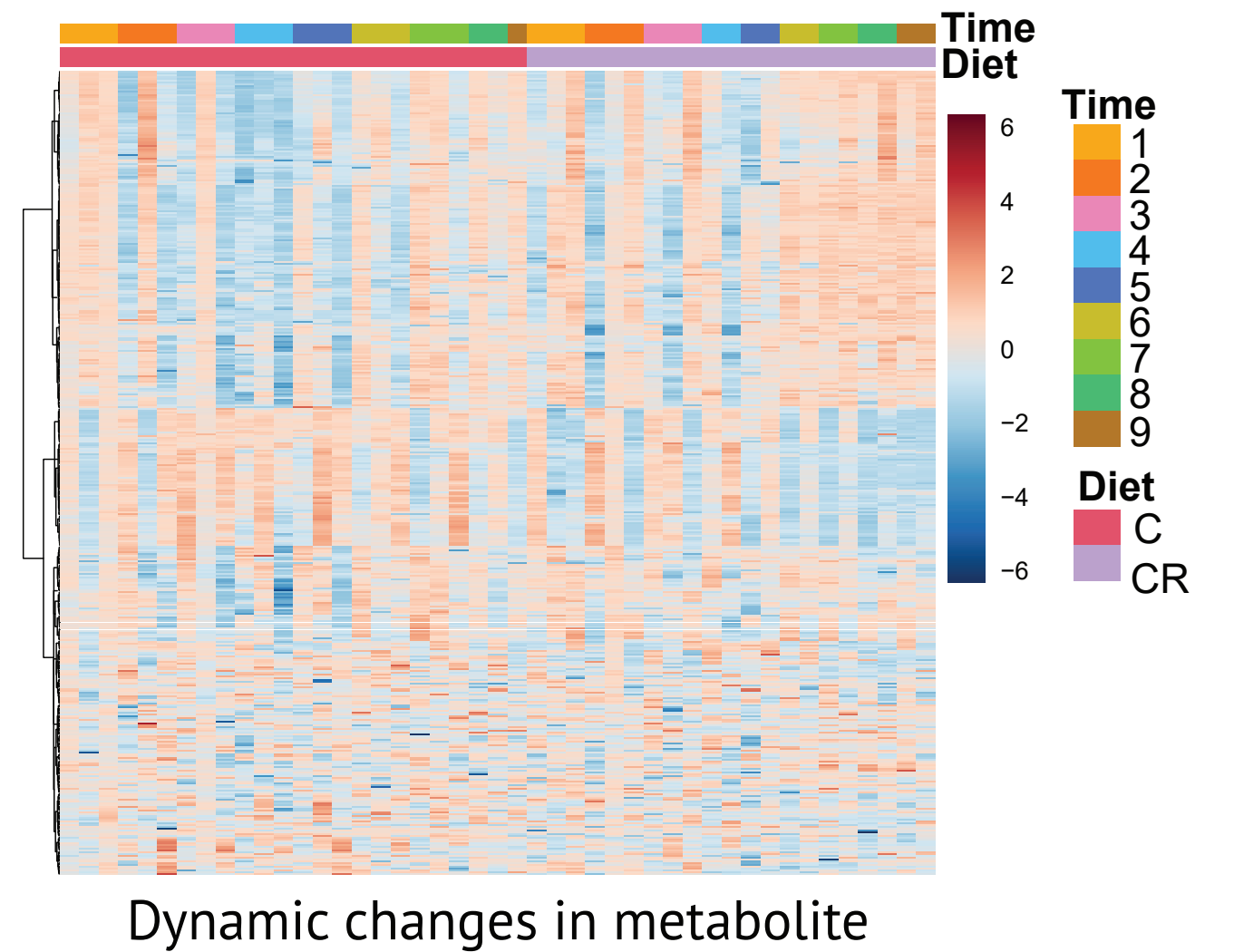
- We successfully identified a total number of 439 metabolites and lipids in all monkey serum samples. The lipid composition revealed a dominant presence of TGs, confirming our previous findings with pooled monkey serum.

- The PCA analysis shows a high degree of overlap between the control and CR groups, indicating shared characteristics in the overall metabolite profiles.
- Alterations in metabolite and lipid expression are observed: TGs were found to be up-regulated in the CR groups, highlighting a potential metabolic shift towards increased lipid utilization. Conversely, PCs were down-regulated, indicating potential changes in lipoproteins.



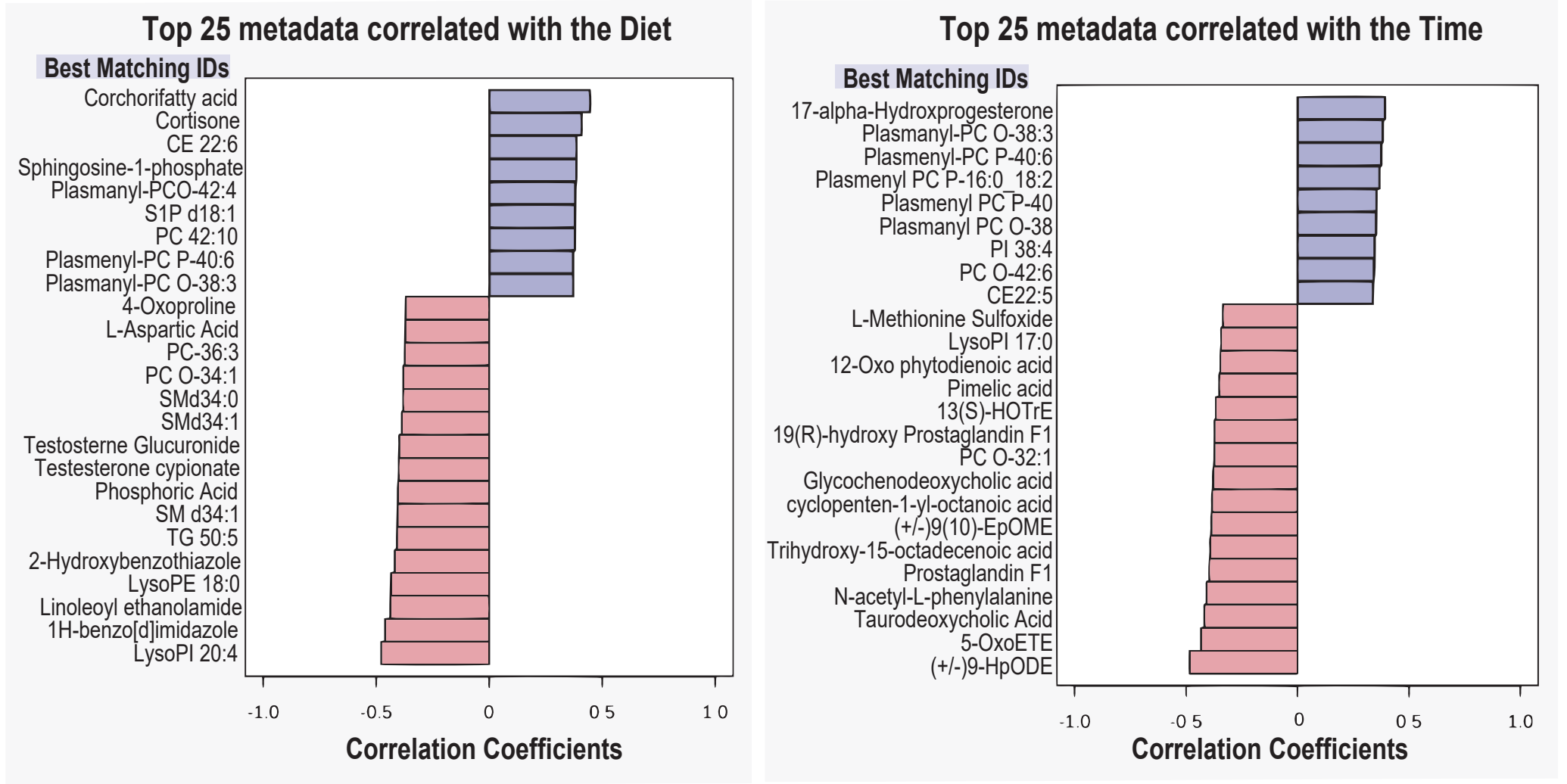
Differential Metabolites and Lipids between CR and Control-Fed Monkeys





Dynamic changes in metabolite expression over time between control and CR groups.

Metabolites and Lipids Correlated with Diet and Age

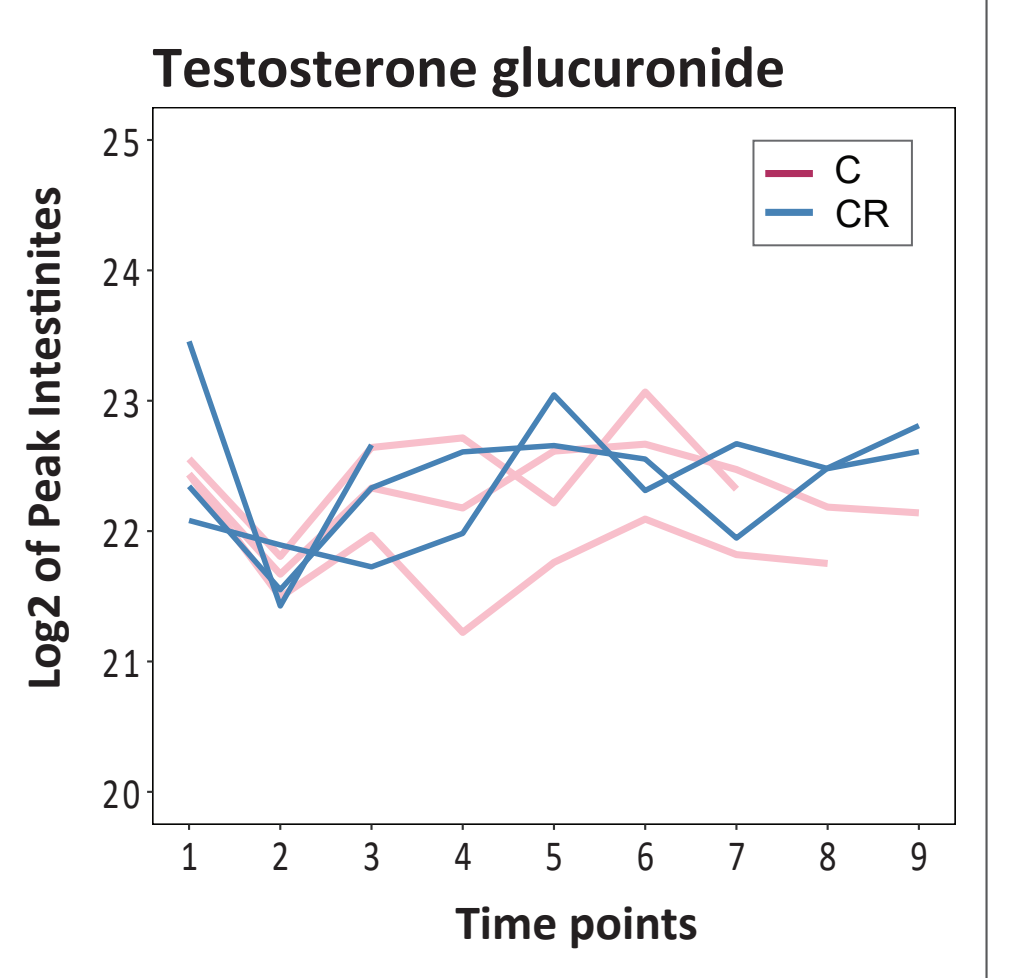


Conclusions

- Our method provides a time-efficient and reproducible approach to provide valuable insights into the benefits of CR on the metabolic landscape.
- Further analyses of the expanded sample sets, including more monkey serum samples, will be needed to provide more comprehensive understanding of the metabolic changes.

Future Directions

- Our data from the correlation analyses suggests a significant increase in testosterone glucuronide levels between the CR group by the end of life compared to early life stages. This finding aligns with previous literature linking higher testosterone levels to improved metabolic health and muscle function.
- Thus, we would like to investigate the life stage-resolved response to CR in NHP and their implication on health throughout time.



Acknowledgments

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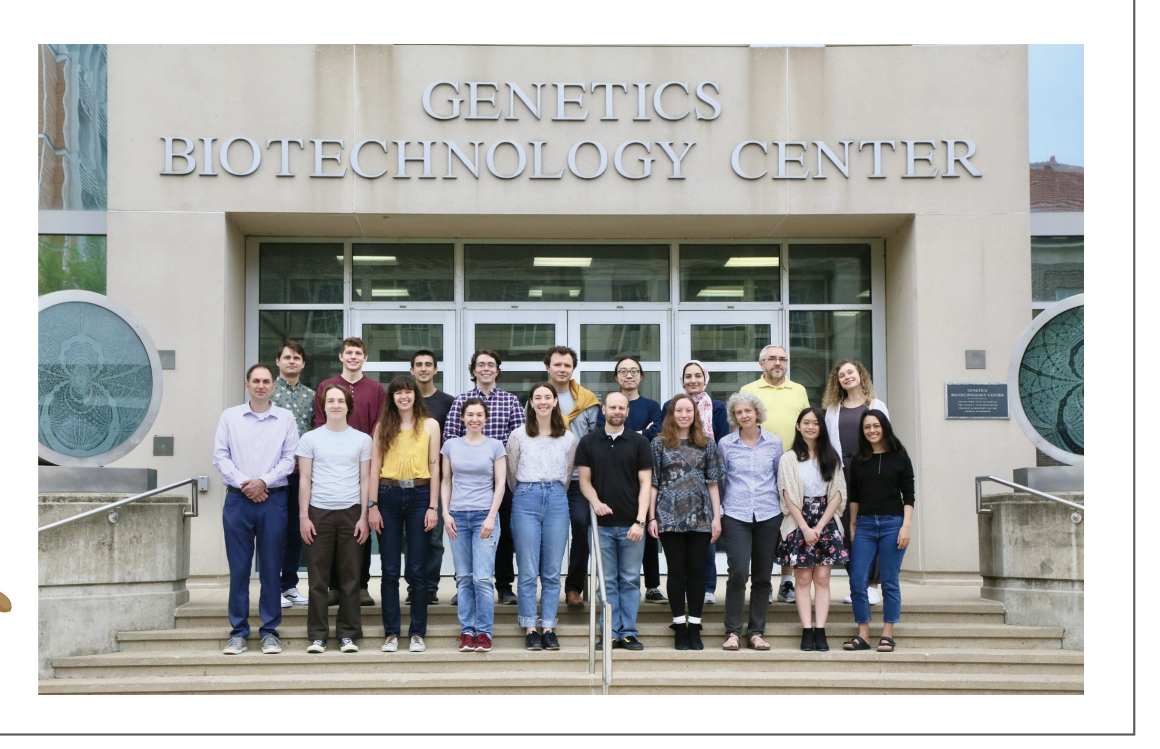
JJC is a consultant for Thermo Fisher Scientific, 908 Devices, and Seer.



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